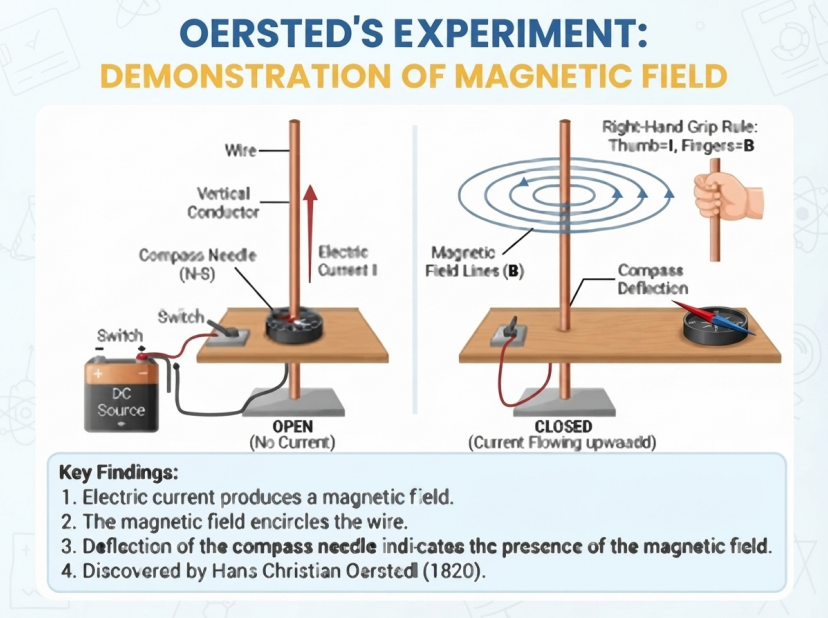
Chapter IV Electromagnetic

In 1819 Oersted accidentally discovered the magnetic effect of an electric current. His experiment can be repeated by holding a wire over and parallel to a compass needle that is pointing N and S (Figure below). The needle moves when the current is switched on. Reversing the current causes the needle to move in the opposite direction. Evidently around a wire carrying a current there is a magnetic field. As with the field due to a permanent magnet, we represent the field due to a current by field lines or lines of force. Arrows on the lines show the direction of the field, i.e. the direction in which a N pole points. Different field patterns are given by differently shaped conductors.



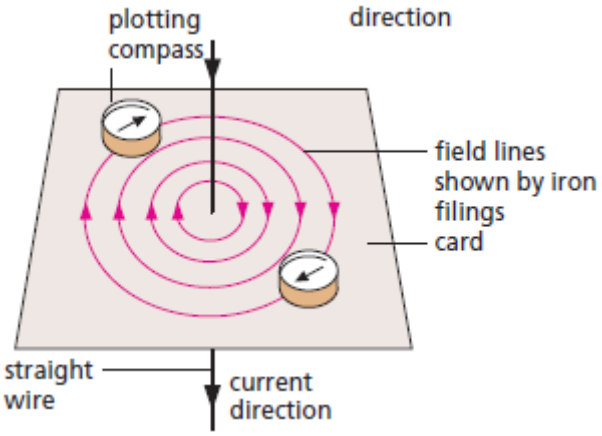
Lesson 1

Magnetic Field

Objective: You will be able to:

- Describe properties of magnetic field
- Determine the direction of magnetic field due to straight wire, solenoid.
- Make electromagnet.

1-The magnetic field due to a straight current-carrying wire



-What happens to the compasses when we place compasses around the current-wire?

Properties of **magnetic field due to a current carrying conductor** can be summarized as below,

-The field lines form concentric circles around the wire.

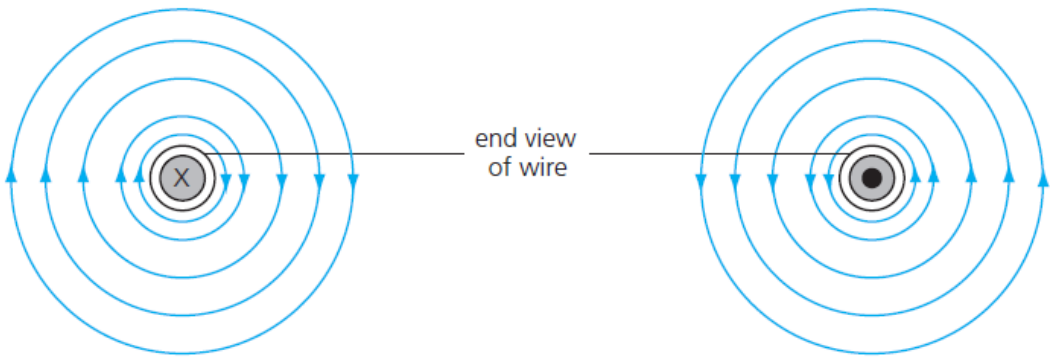
- The direction of the magnetic field is perpendicular to the wire.

-The direction of the field is found using right hand rule

☞*place your thumb in direction of current*

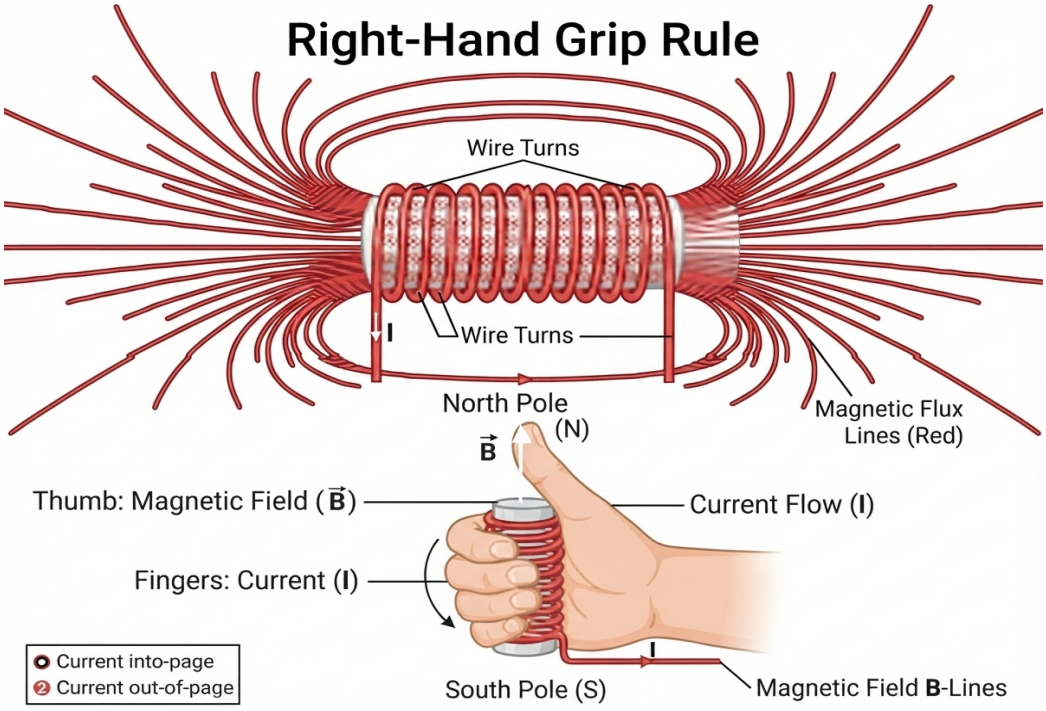
☞Y*our 4 fingers that wrapped around the wire will point to the direction of magnetic field*).

The right-hand grip rule will help you to remember this. Put the thumb of your right hand along a wire in the direction of the current. Now your fingers point in the direction of the magnetic field.



2-The magnetic field lines in Solenoid

Solenoid is a long, thin loop of wire, which when wrapped around a piece of metal, produces a magnetic field when the current passes through it.



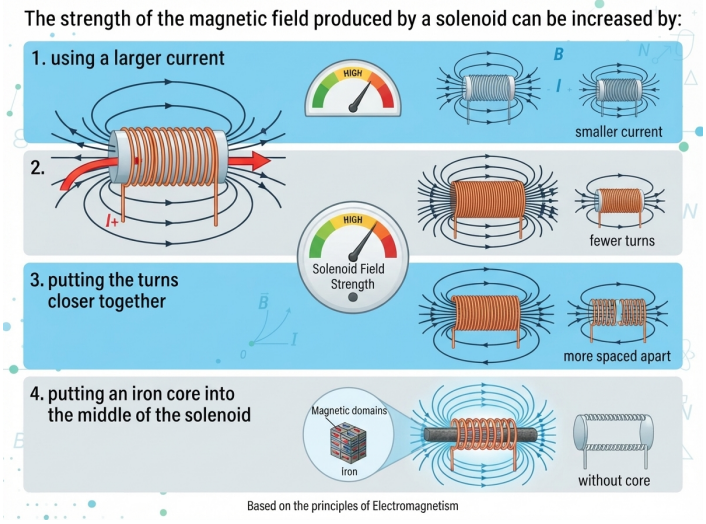
Right-hand grip rule for solenoid:

-If the **right hand grips** the solenoid so that the 4 fingers curl in the same way as the current

- the thumb will point the same way as the field lines inside the solenoid.

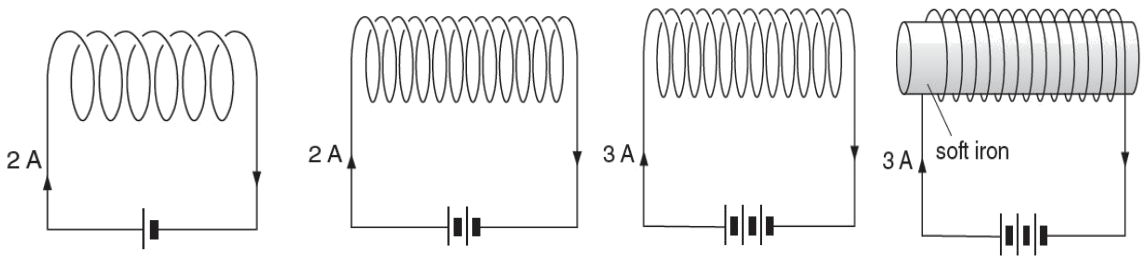
The strength of the magnetic field produced by a solenoid can be increased by:

- using a larger current
- using more turns of wire
- putting the turns closer together
- putting an iron core into the middle of the solenoid.



The **magnetism** of the soft-iron core dies away as soon as the current is switched off. If the core is made of steel instead, it retains its magnetism even after the current has been switched off.

Example: Which of the following coils gives the strongest magnetic field?



3-Electromagnets



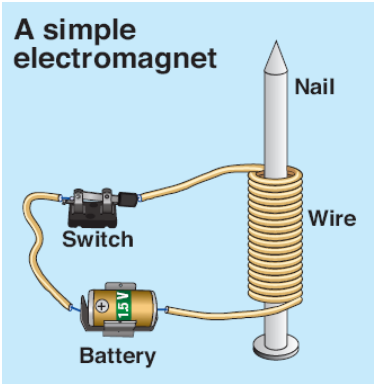
An **electromagnet** is a type of magnet in which the magnetic field is produced by the flow of electric current.

An electromagnet is made from a coil of wire that acts as a magnet when an electric current passes through it but stops being a magnet when the current stops. Often, the coil is wrapped around a core of ferromagnetic material like steel, which enhances the magnetic field produced by the coil.



Electromagnets are very widely used in electric and electro-mechanic devices, including:

- Motors and generators
- Transformers
- Electric bells
- Loudspeakers



- Magnetic recording and data storage equipment: tape recorders, hard disks.
- Scientific instruments.
- Magnetic locks.
- Industrial lifting magnets.

The magnetic field inside an electromagnet can be made stronger by:

• Increasing the number of turns on the coil.

• Increasing the current through the coil.

• Inserting a soft-iron core into the coil.

Applications of Electromagnets

Aluminum diaphragm vibrates when you speak into the mouthpiece. Compresses the carbon granules repeatedly, Varying current to electromagnet in earpiece. Iron diaphragm vibrates and transmits sound waves into your ear.

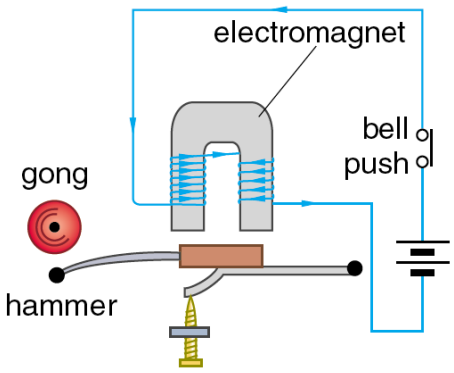
Scrap yard crane

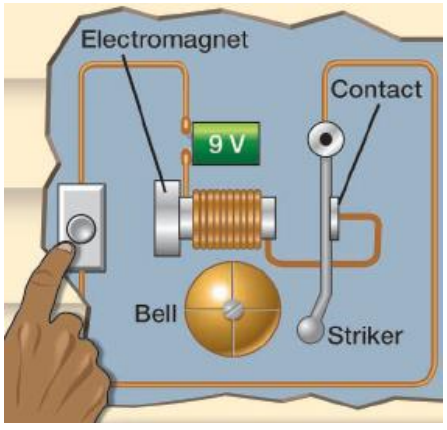
When current flows the iron becomes strongly magnetised and so picks up the scrap iron and steel.

When the current is turned off the iron loses its magnetisation and so releases the scrap.

A door bell

- A doorbell contains an electromagnet.
- When the button of the bell is pushed, it sends current through the electromagnet.





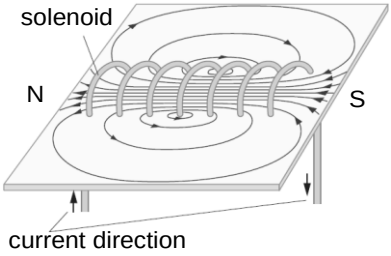
Questions-Exercises

1- From the field pattern of the solenoid (Fig 4l), it can be found that:

i. inside the solenoid, field lines are _____and_____

⇒ uniform field

ii.outside the solenoid, the pattern is similar to that of a bar magnet.



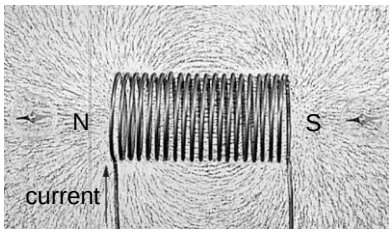


Fig 4l

2- The poles of the solenoid can be worked out using the _____ for solenoid (Fig 4m).

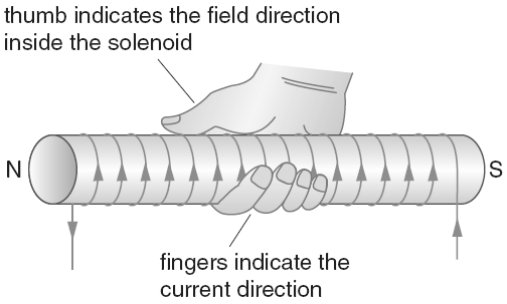
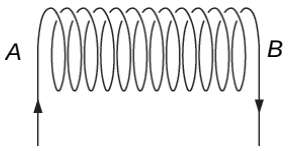


Fig 4m

If the right hand grips the solenoid so that the fingers curl in the same way as the_____, the thumb points the same way as the _____ inside the solenoid.

3-A solenoid has a current in the direction shown.

Identify the magnetic poles at the end *A* and *B*.



4- The B-field of the solenoid can be increased by

i. _____ the current,

ii. _____the no. of turns in the solenoid (for the same length of solenoid),

iii. Inserting a _____ through the solenoid

5- The magnetism of the _____ core dies away as soon as the current is switched off. If

the core is made of steel instead, it _____ its magnetism even after the current has been switched off.

6-The ends of a solenoid behave like a N-pole and a S-pole when a current passes through the solenoid (Fig a).

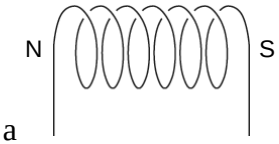


Fig a

(a) Indicate the direction of the current in Figure a.

(b) What happens to the magnetic field produced if,

(i) the number of turns decreases?

(ii) an aluminium core is placed inside the solenoid?

(c) What happen to the magnetism of the solenoid after the current is switched off?

7-An electromagnet can be made strong by

i- _____the number of turn on the coil,

ii- _____the current through the coil,

iii- inserting a_____ into the coil.